Exponentiation

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1095 | Mathematics | https://cses.fi/problemset/task/1095

Your task is to efficiently calculate values a^b modulo 10^9+7.

Note that in this task we assume that $0^0=1$.

Input

The first input line contains an integer n: the number of calculations.

After this, there are n lines, each containing two integers a and b.

Output

Print each value a^b modulo 10^9+7.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $0 \le a,b \le 10^9$

Example

Input

```
3
3 4
2 8
123 123
```

Output

```
81
256
921450052
```